

Q1) Fill the table with the correct answer (12 marks)

1	2	3	4	5	6	7	8	9	10	11	12

1) Given that register CH contains an unsigned number N. What are the minimum and maximum possible values of N, respectively?

- A) 0 and 127. B) 256 and 255. C) -128 and 128. D) 0 and 255

2) If 1110 0100 is the 8-bit 2's complement representation of a number N, what is the decimal value of N?

- A) -28 B) +228 C) -228 D) -74

3) Which of the following summation set the CF and SF?

- A) 7FF1h + 000Fh B) 7FFEh + 0001h C) FFO0h + FF00h D) B000h + B000h

4) Which of the following statement is correct, if X DB 22H

- A) MOV [SP], 99H B) MOV BL, OFFSET X C) PUSH XD) none of them

5) What is the type of addressing for the following statement: MOV BX, [SI]

- A) Indexed B) indexed based C) direct D) based

6) If you want to compare the following two signed values AX = 123Fh, BX = B3F0h and put the value 1 in CL if AX < BX, you write:

- A) CMP AX, BX JL L1 B) CMP AX, BX JA L1 C) CMP AX, BX JB L1 D) CMP AX, BX JG L1

L1: MOV CL, 1 L1: MOV CL, 1 L1: MOV CL, 1 L1: MOV CL, 1

7) The value of CX after executing the following code is:

MOV AL, 0FFH

MOV CX, 5

L1: ADD AL, 1

LOOPZ L1

- A) CX=0 B) CX=5 C) CX= 4 D) CX=3

8) What is the value of the AL register after executing the following code?

```
MOV AX, 0FF02H
MOV CL, 0FCH
XCHG AL, CL
ADD AL, 5
SUB AL, 1
```

- A) 01H B) 00H C) FFH D) None of them

9) Number of the times the instruction sequence below will loop before coming out of loop is

```
MOV AL, 00H
L1: INC AL
JNZ L1
```

- A) 0 B) 1 C) 256 D) Infinite Loop

Assume: SS=1000H, DS=7000H, SI=0020H, DI=0100H, X=0500H, BP=0300h

10) What is the physical address of value that will be stored in DH for the following instruction `MOV DX, X[SI+2]`

- A) 7522h B) 10522h C) 70522h D) 70523h

11) Calculate the physical address of the destination operand for the following instruction `MOV [BP + 40H], CL`

- A) 70340h B) 10340h C) 1340h D) 7340h

12) What is the value of the SP register after executing the following instruction, if the SP = FFFAH? `POP AX`

```
DEC SP
PUSH BX
```

- A) FFF7H B) FFF8H C) FFFBH D) FFF9H

Q2) Trace the following codes and find the result of executing by filling the blanks. (25 marks)

```
1) MOV CX, 8
   MOV BL, 2
   MOV AL, 9
L: MOV AH, 0
   DIV BL
   CMP AL, 0
   LOOPNZ L
```

AH=

AL=

BL=

CX=

```

2) MOV AX, 008AH
   MOV BX, F380H
   MOV BL, 2
   ADD AL, DL
   CMP DH, 2
   JL L
   DIV BL
   JMP L1
L: MUL BL
L1: HLT

```

AH=

AL=

DH=

DL=

```

3) X DW 733H, 1122H
   MOV AX, 4455H
   PUSH AX
   PUSH X+1
   POP AX
   POP BX

```

AH=

AL=

BH=

BL=

```

4) X DB 3, 5, 7
   INC X+1
   MOV AX, WORD PTR X+1
   NEG AX

```

AL=

AH=

CF=

SF=

PF=

AF=

```

5) X DW 5437H
   Y DW 28H
   MOV BX, OFFSET X
   ADD BX, 1
   MOV AL, [BX]
   MOV CL, [BX+1]
   ADD AL, CL
   INC BX
   MOV DL, [BX]

```

DL=

AH=

AL=

CL=

```

6) X DB 3, 5, 4
   MOV CX, 2
   MOV SI, 1
L2: TEST X[SI], 81H
   JZ L1
   NOT [SI]
L1: INC SI
   LOOP L2

```

X = _____, _____, _____

Q3)

a) Suppose we have a microprocessor with 24 bits address bus, what is the maximum memory capacity will be? (1 mark)

b) Write the equivalent assembly codes to the following C++ codes: (3 marks)

Note: x, y and s are unsigned byte variable.

```

if( x > y)
    s = s + x;
else
    s = s - y;

```

Q#1:[12] Circle the correct answer:

1. Data bus width of Intel 8086 microprocessor is

- (a) 16 (b) 24 (c) 20 (d) 32

2. The length of extended ASCII codes

- (a) 7 bits (b) 8 bits (c) 12 bits (d) 16 bits

3. One of the following is not part of execution unit of 8086 microprocessor

- (a) BP register (b) AX register (c) DS register (d) SP register

4. The equivalent code of NEG AX instruction is:

- (a) XOR AX,AX
SUB AX,1 (b) SUB AX,AX
ADD AX,1 (c) OR AX,AX
ADD AX,1 (d) XOR AX,FFFFH
ADD AX,1

5. A 16-bit address bus allows access to a memory of capacity

- (a) 128KB (b) 64KB (c) 256KB (d) 512KB

6. If BX=8321h, CF=1 and CL=5 after execution ROR BX,CL the:

- (a) BX=0C19H and CF=1 (b) BX=0C19H and CF=0
(c) BX=1832H and CF=0 (d) BX=1832H and CF=1

7. One of the following control transfer instruction is not used with signed number

- (a) JGE (b) BE (c) JL (d) JLE

8. After execution of the following code register AL contains

```
MOV AL, 'a'
ADD AL, 2
AND AL, 1101 1111b
```

- (a) AL='c' (b) AL='C' (c) AL='a' (d) AL='A'

10. If the SP register contains the decimal value 340, what will be its value after executing the following program segment?

```
PUSH AX      MOV BX, 25
PUSH BX      MUL BX
PUSH DX      POP BX
```

- a) 344 b) 338 c) 342 d) 336

11. Given that register CH contains an unsigned number N. What are the minimum and maximum possible values of N, respectively?

- a) 0 and 127, b) 256 and 255
c) -128 and 128, d) 0 and 255

12. If 1110 0100 is the 8-bit 2's complement representation of a number N, what is the decimal value of N?

- a) -28, b) +228,
c) -228, d) -74,

Q2: [15 points] Trace the following programs and determine the final value for each register:

a) MOV AL, 05H
NEG AL
MOV BL, 02H
AND BL, AL

BL=	AL=	PF
2	FB	0

b) MOV BL, 07H
SUB BL, 0EH
MOV DL, 02H
MOV AL, FFH
ADD AL, 1
JCL
MOV CL, FFH
JNC LL
L: MOV CL, 00
LL: ADC DL, 1

AL=	BL=	CL=	DL=
00	F9	00	02

c) MOV CL, 03H
MOV AX, 127
ROR AX, 01H
JCL
AND CL, 02H
L: HLT

AX=	CL=	CF=	ZF=	PF=
803F	03	1	0	1

a) MOV AX, 35H
 MOV BX, 39H
 PUSH AX
 PUSH BX
 ADD AX, BX
 DAA
 MOV DX, AX
 POP AX
 POP BX

AX=	BX=	DX=
35	39	34

b) MOV AL, 14
 MOV AH, 0
 MOV BL, 17
 DIV BL

AL = 0.82 AH = 00

Question #2

Suppose we have a microprocessor with 24 bits address bus:

a) What is the maximum memory capacity will be?

$\text{max memory} = 2^{24}$

b) For the following instruction MOV [BX+76Dh], AL

1- What is the addressing mode used?

Base addressing mode with displacement

2- Calculate the physical address of memory location assuming that DS=8CFBh and BX=4000h

ph address = reg * 10h + off

[5 points]

phys add = 8CFBh * 10h + 4000h
 = 90FB0

Question #3

Write an assembly code to clear low order 4 bit of memory locations and store the result into Y. [4 points]

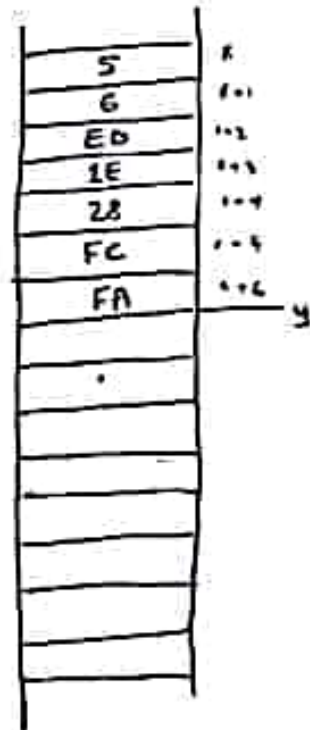
X DB 5,6,-20,30,40,1,-1,-6
Y DW 8 DUP(0)

MOV AH,[X]

~~MOV AH,[X]~~
LOOP

MOV Y,AH

HLT



Question #4

Given 256 KB of memory locations, Write an assembly code to find the sum of memory locations that values greater than 5. [4 points]